Physics 509C Assignment 1

1. $P_1(H|1) = 0.5$

$P_2(H|2) = 0.9$

$P(1) = P(2) = 0.5$

$$P_1(H) = P_1(H|1)\,P(1) + P(H|2)\,P(2)$$

$$= 0.5 \cdot 0.5 + 0.9 \cdot 0.5$$

$$= 0.7$$

$$\therefore P_1(T) = 1 - P_1(H) = 0.3$$

$$P_2(HH) = \left(P_1(H|1)\right)^2 P(1) + \left(P_1(H|2)\right)^2 P(2)$$

$$= (0.5)^2 \cdot 0.5 + (0.9)^2 \cdot 0.5$$

$$= 0.53$$

since order does not matter

$$P_2(HT) = P(TH) = P_1(H|1)\,P_1(T|1)\,P(1) + P_1(H|2)\,P(T|2)\,P(2)$$

$$= 0.5^3 + 0.9 \cdot 0.1 \cdot 0.5$$

$$= 0.17$$

$$P_2(TT) = \left(P_1(T|1)\right)^2 P(1) + \left(P_1(T|2)\right)^2 P(2)$$

$$= 0.5^3 + 0.1^2 \cdot 0.5$$

$$= 0.13$$

These add up to 1 (as they should).

$P_2(HT) = 0.17$

$P_1(H)\,P_1(T) = 0.7 \cdot 0.3 = 0.21$

$\therefore P_2(HT) \neq P_1(H)\,P_1(T)$

$$P(H|T) = \frac{P_2(HT)}{P_1(T)} = \frac{0.17}{0.3} = 0.57$$

$$P(T|H) = \frac{P_2(TH)}{P_1(H)} = \frac{0.17}{0.7} = 0.24$$

2. $P_x(x) = \frac{1}{\sqrt{2\pi}\,\sigma_x} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}}$ and similarly for y.

$$P_z(z) = \int_{-\infty}^{\infty} dx \frac{1}{\sqrt{2\pi}\,\sigma_x} e^{\frac{-(x-\mu_x)^2}{2\sigma_x^2}} \frac{1}{\sqrt{2\pi}\,\sigma_y} e^{\frac{-(z-x-\mu_y)^2}{2\sigma_y^2}}$$

From here you need to multiply out & complete the square but i'm just writing & don't want to do all the math for this sorry.

4. a) In the N trials there must be m successes. The probability of this is p^m. There must also be $N-m$ failures. The probability of this is $(1-p)^{N-m}$. Now, we need to account for how the failures are distributed among the N trials. The Nth trial <u>must</u> be a success so that leaves $m-1$ successes to be distributed among $N-1$ trials. There are $\binom{N-1}{m-1}$ ways to do this.

$$\therefore P(N \mid m, p) = \binom{N-1}{m-1} p^m (1-p)^{N-m}$$

assignment1

September 24, 2026

1 Assignment 1

[98]:
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.special import binom
```

1.1 Question 3

[ ]:
```
def est_1(vals,mu):
    '''
    Estimator 1.

    Parameters:
    ----------
    vals : input values x_i
    mu : expectation value
    '''
    return np.sum((vals-mu)**2)/np.shape(vals)[0]

def est_2(vals):
    '''
    Estimator 2.

    Parameters:
    ----------
    vals : input values x_i
    '''
    xbar = np.mean(vals)
    return np.sum((vals-xbar)**2)/np.shape(vals)[0]

def est_3(vals):
    '''
    Estimator 3.

    Parameters:
    ----------
    vals : input values x_i
```

```
    '''
    xbar = np.mean(vals)
    return np.sum((vals-xbar)**2)/(np.shape(vals)[0]-1)
```

[90]:
```
# draw 10 numbers from gaussian distribution
mu = 2
sigma = 1
nums = np.random.normal(mu,sigma,(10,))
nums
```

[90]:
```
array([2.8125311 , 2.72157498, 1.43183061, 1.70366828, 3.45478352,
       1.12793118, 1.06937779, 1.80203546, 1.1716036 , 0.71892151])
```

[91]:
```
# print the output from the three estimators
print(est_1(nums,mu),est_2(nums),est_3(nums))
```

```
0.7701055649873388 0.7306738530962105 0.8118598367735672
```

[93]:
```
# repeat this for M trials
M = 100
est_1_vals = []
est_2_vals = []
est_3_vals = []
for m in range(M):
    nums = np.random.normal(mu,sigma,(10,))
    est_1_vals.append(est_1(nums,mu))
    est_2_vals.append(est_2(nums))
    est_3_vals.append(est_3(nums))

print(np.mean(est_1_vals),np.mean(est_2_vals),np.mean(est_3_vals))
```

```
0.9945862153605423 0.8921062044346222 0.9912291160384692
```

Repeating this for 100 trials makes it clear that the second estimator is definitely biased small.

1.2 Question 4

[95]:
```
def neg_bin_dist(N,m,p):
    '''
    Negative binomial ditribution.

    Parameters:
    ----------
    N : number of trials
    m : number of successes
    p : probability of success per trial
    '''
    return binom(N-1,m-1) * p**m * (1-p)**(N-m)
```

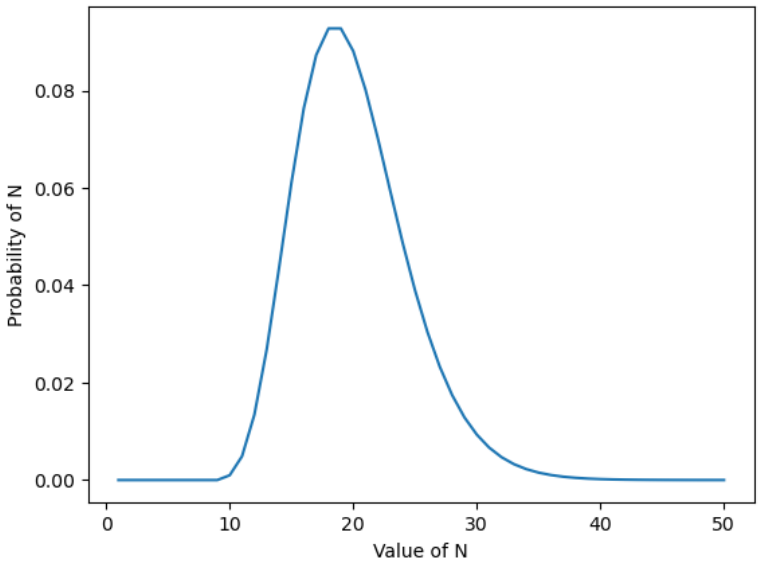
[221]:
```
# plot values of N between 1 and 50
N = np.linspace(1,50,50)
```

[229]:
```
fig,ax = plt.subplots(1,1)
ax.plot(N,neg_bin_dist(N,10,0.5))

ax.set_xlabel('Value of N')
ax.set_ylabel('Probability of N')

caption = 'Distribution for the number of trials $N$ required to obtain $m=10$␣
 ↪heads in a fair coin toss ($p$ = 0.5)'
plt.figtext(0.5, -0.05, caption, wrap=True, horizontalalignment='center',␣
 ↪fontsize=12)
```

[229]:
```
Text(0.5, -0.05, 'Distribution for the number of trials $N$ required to obtain
$m=10$ heads in a fair coin toss ($p$ = 0.5)')
```



Distribution for the number of trials N required to obtain $m = 10$ heads in a fair coin toss (p = 0.5)

This is not symmetric about the mode.

1.3 Question 5

[119]:
```
def post_density(x,y):
    '''
    '''
    A1 = 3.93161
    A2 = 3.61472
    x1 = 0.5
```

```
    x2 = 0.65
    y1 = 0.5
    y2 = 0.75
    sig1 = 0.2
    sig2 = 0.04
    return A1 * np.exp(-(((x-x1)**2+(y-y1)**2)/(2*sig1**2))) + A2 * np.
↪exp(-(((x-x2)**2+(y-y2)**2)/(2*sig2**2)))
```

[166]:
```
# make x and y arrays
x = np.linspace(0,1,100)
y = np.linspace(0,1,100)
```

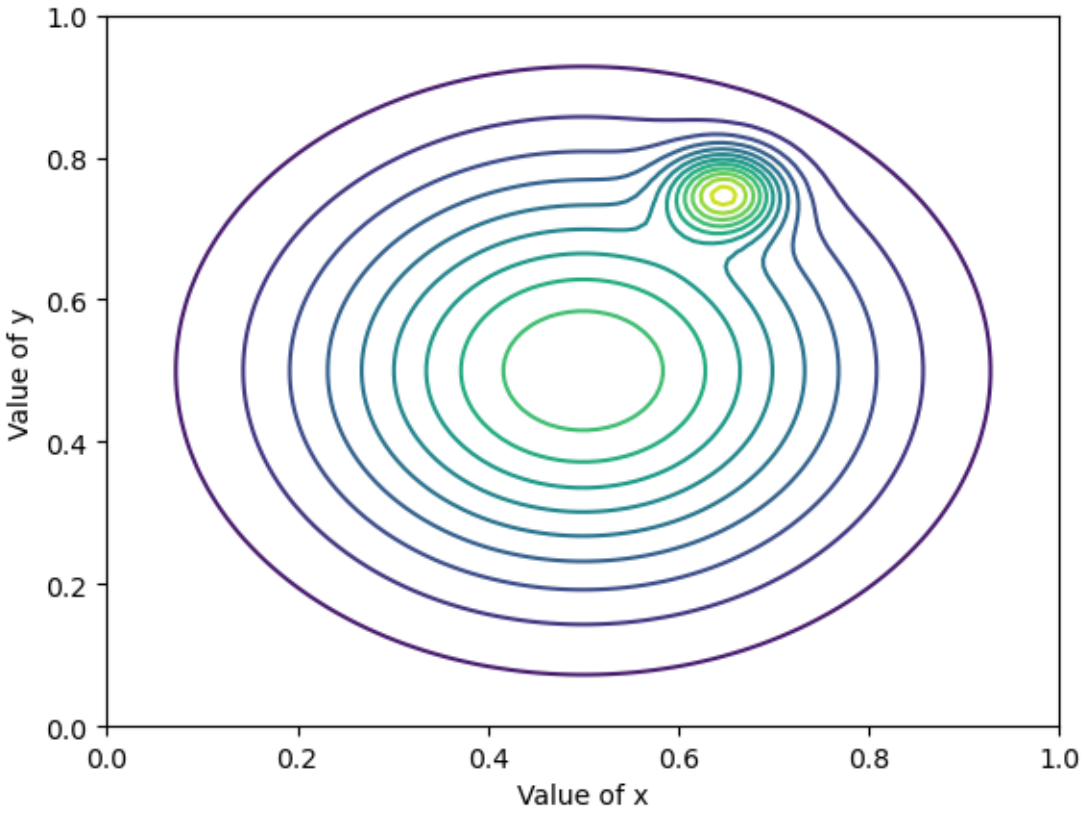
[165]:
```
xplt = np.meshgrid(x,y)[0]
yplt = np.meshgrid(x,y)[1]
```

1.3.1 Part a)

[230]:
```
fig,ax = plt.subplots(1,1)
ax.contour(xplt, yplt, post_density(xplt,yplt),levels=15)
ax.set_xlabel('Value of x')
ax.set_ylabel('Value of y')

caption = 'Contour plot of posterior density.'
plt.figtext(0.5, -0.05, caption, wrap=True, horizontalalignment='center',␣
↪fontsize=12)
```

[230]:
```
Text(0.5, -0.05, 'Contour plot of posterior density.')
```



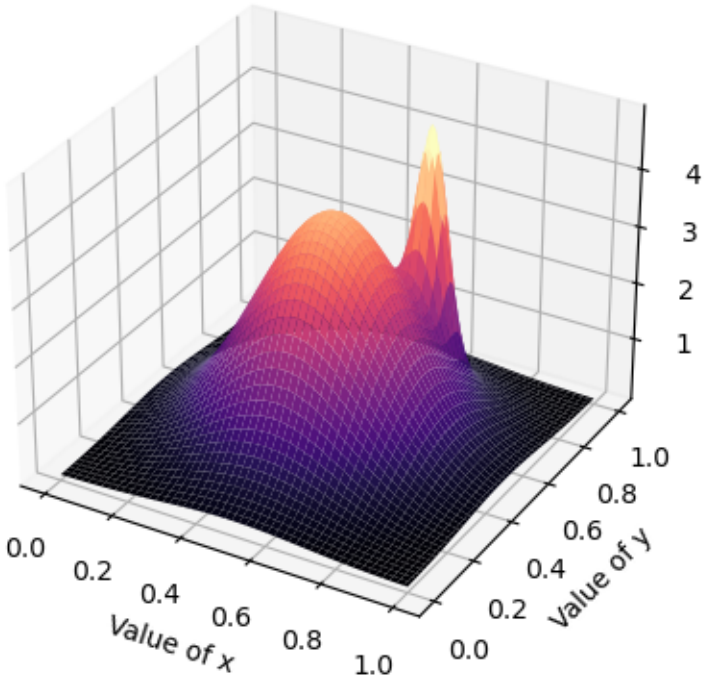
Contour plot of posterior density.

1.3.2 Part b)

[232]:
```
fig, ax = plt.subplots(subplot_kw=dict(projection='3d'))
ax.plot_surface(xplt, yplt, post_density(xplt,yplt),cmap='magma')
ax.set_xlabel('Value of x')
ax.set_ylabel('Value of y')

caption = '3D plot of posterior density.'
plt.figtext(0.5, -0.05, caption, wrap=True, horizontalalignment='center',␣
↪fontsize=12)
```

[232]:
```
Text(0.5, -0.05, '3D plot of posterior density.')
```



3D plot of posterior density.

1.3.3 Part c)

[161]:
```
ints = np.linspace(0,1,1001)[:-1]
dx = np.diff(ints)[0]
dy = dx
```

[234]:
```
fig,axs = plt.subplots(1,2)

xpltint = np.meshgrid(x,ints)[0]
ypltint = np.meshgrid(x,ints)[1]

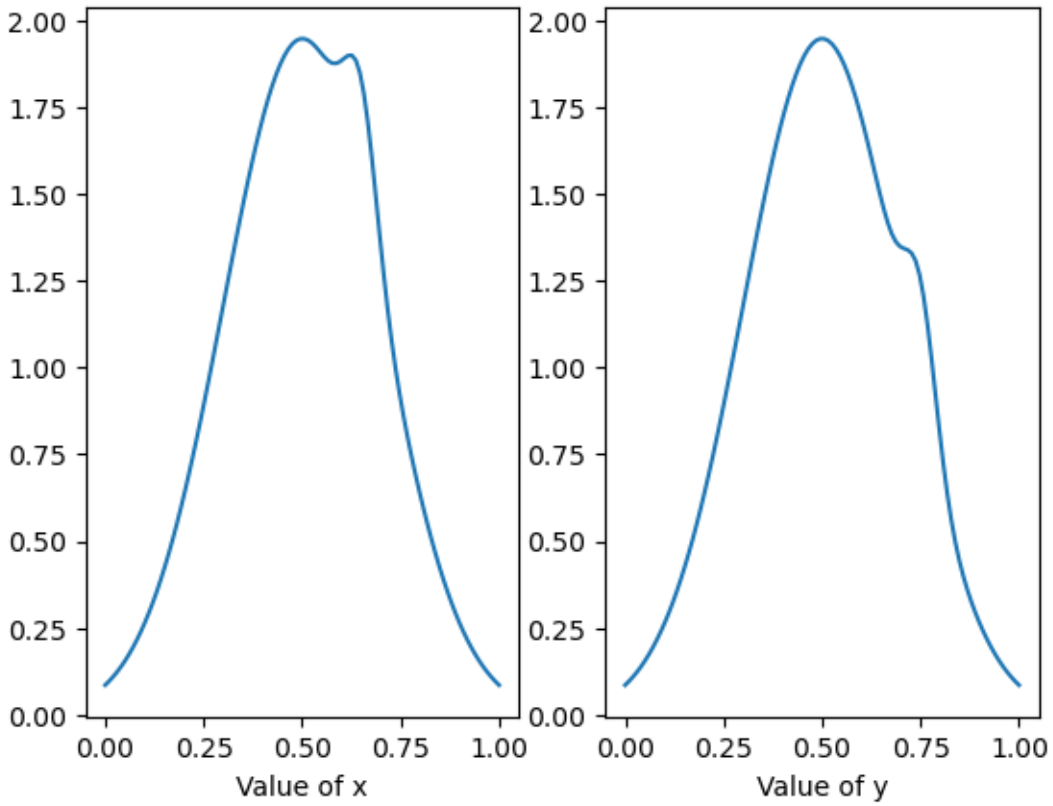
axs[0].plot(x,np.sum(post_density(xpltint,ypltint).T,axis=1)*dy)
axs[0].set_xlabel('Value of x')

xpltint = np.meshgrid(ints,y)[0]
ypltint = np.meshgrid(ints,y)[1]

axs[1].plot(y,np.sum(post_density(xpltint,ypltint),axis=1)*dx)
axs[1].set_xlabel('Value of y')
```

```
caption = '$x$ and $y$ marginal probability distributions for posterior density.
 ↪'
plt.figtext(0.5, -0.05, caption, wrap=True, horizontalalignment='center',␣
 ↪fontsize=12)
```

[234]:
```
Text(0.5, -0.05, '$x$ and $y$ marginal probability distributions for posterior
density.')
```



x and *y* marginal probability distributions for posterior density.

[172]:
```
# check to make sure it integrates to 1
xpltint = np.meshgrid(ints,ints)[0]
ypltint = np.meshgrid(ints,ints)[1]
np.sum(np.sum(post_density(xpltint,ypltint).T,axis=1)*dy)*dx
```

[172]:
```
1.0000689356551917
```

1.3.4 Part d)

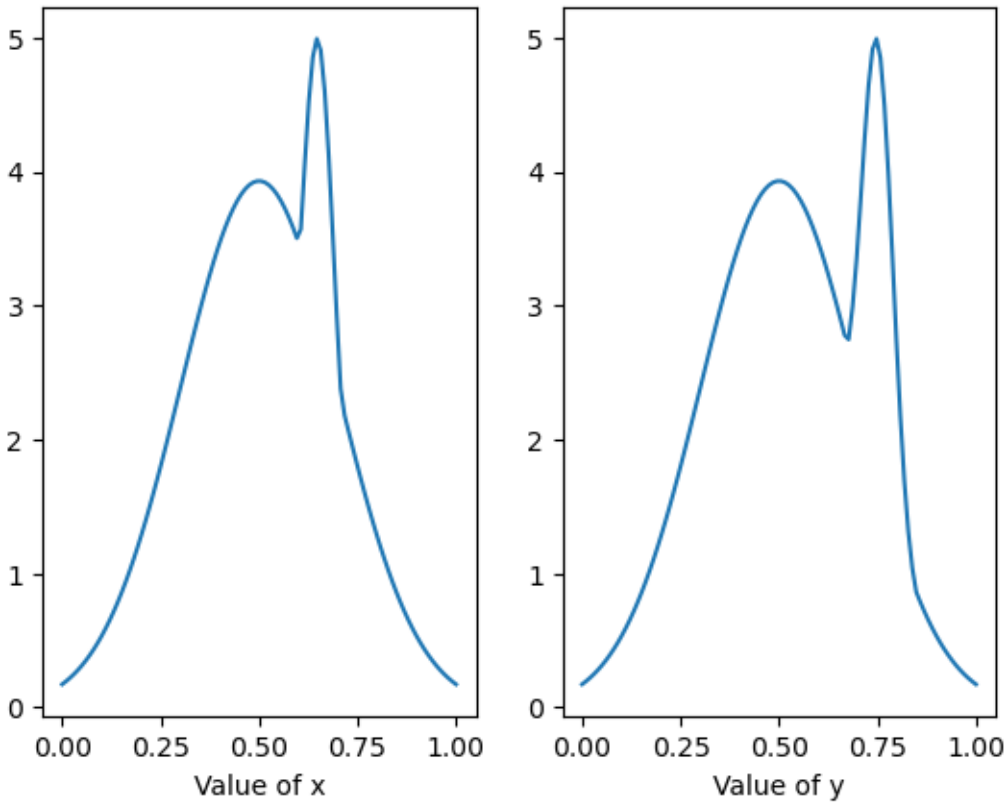
[235]:
```
fig,axs = plt.subplots(1,2)

axs[0].plot(x,np.max(post_density(xplt,yplt).T,axis=1))
axs[0].set_xlabel('Value of x')

axs[1].plot(y,np.max(post_density(xplt,yplt),axis=1))
axs[1].set_xlabel('Value of y')

caption = '$x$ and $y$ projected probability distributions for posterior␣
 ↪density.'
plt.figtext(0.5, -0.05, caption, wrap=True, horizontalalignment='center',␣
 ↪fontsize=12)
```

[235]:
```
Text(0.5, -0.05, '$x$ and $y$ projected probability distributions for posterior
density.')
```



x and *y* projected probability distributions for posterior density.

1.3.5 Part e)

[236]:
```
fig,axs = plt.subplots(1,2)

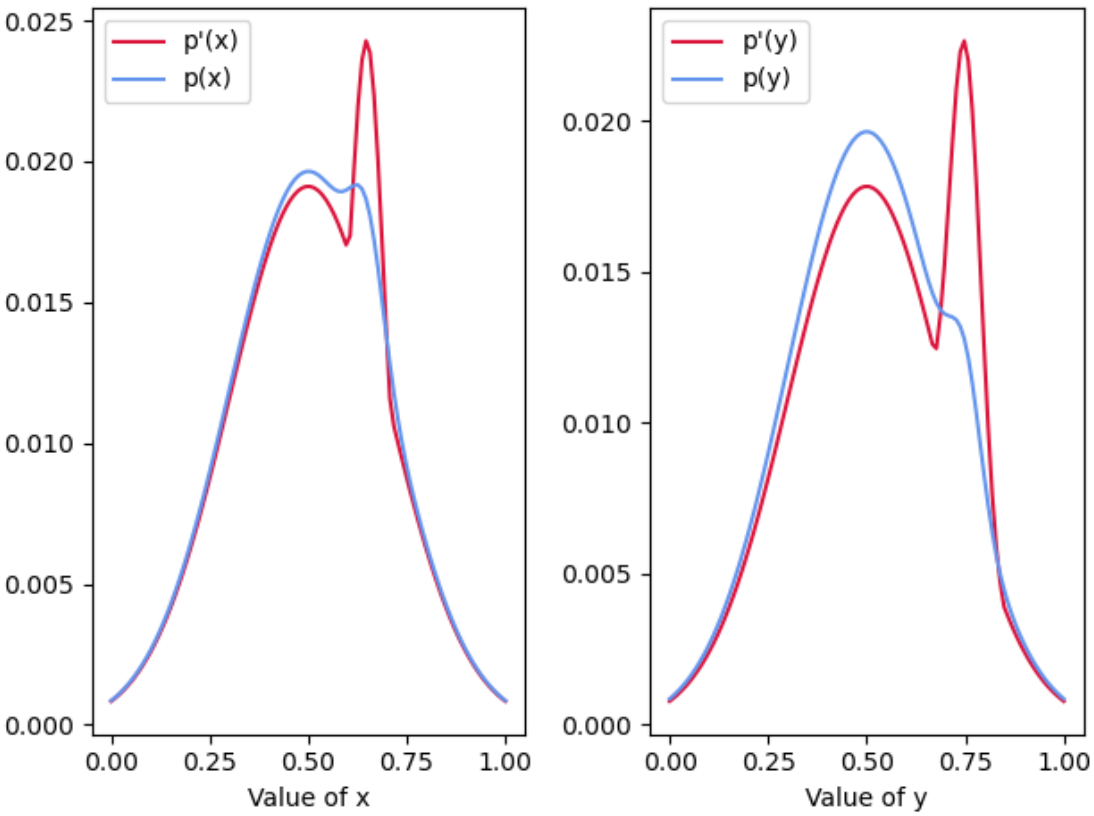
xpltint = np.meshgrid(x,ints)[0]
ypltint = np.meshgrid(x,ints)[1]
axs[0].plot(x,np.max(post_density(xplt,yplt).T,axis=1)/np.sum(np.
 ↪max(post_density(xplt,yplt).T,axis=1)),label='p\'(x)',color='crimson')
axs[0].plot(x,np.sum(post_density(xpltint,ypltint).T,axis=1)*dx/np.sum(np.
 ↪sum(post_density(xpltint,ypltint).
 ↪T,axis=1)*dx),label='p(x)',color='cornflowerblue')
axs[0].set_xlabel('Value of x')
axs[0].legend()

xpltint = np.meshgrid(ints,y)[0]
ypltint = np.meshgrid(ints,y)[1]
axs[1].plot(y,np.max(post_density(xplt,yplt),axis=1)/np.sum(np.
 ↪max(post_density(xplt,yplt),axis=1)),label='p\'(y)',color='crimson')
axs[1].plot(y,np.sum(post_density(xpltint,ypltint),axis=1)*dx/np.sum(np.
 ↪sum(post_density(xpltint,ypltint),axis=1)*dy),label='p(y)',color='cornflowerblue')
axs[1].set_xlabel('Value of y')
axs[1].legend()

fig.tight_layout()

caption = 'Comparison of $x$ and $y$ marginal and projected probability␣
 ↪distributions.'
plt.figtext(0.5, -0.05, caption, wrap=True, horizontalalignment='center',␣
 ↪fontsize=12)
```

[236]:
```
Text(0.5, -0.05, 'Comparison of $x$ and $y$ marginal and projected probability
distributions.')
```



Comparison of x and y marginal and projected probability distributions.